Quiz 6 — Solutions

M 375T Mathematical Foundations of Machine Learning — Fall 2026 · Thursday, October 15

Equilibrium and detailed balance (10 points)

(a) Invariant measure: $P^{\mathsf{T}}\pi = \pi$, i.e. $\sum_i \pi_i P_{ij} = \pi_j$ for every j. What is balanced: the total mass at each vertex — what arrives at j in one step equals what was there, so the law does not change. Detailed balance: $\pi_i P_{ij} = \pi_j P_{ji}$ for every pair i, j. What is balanced: the traffic on each edge — the mass going from i to j equals the mass going from j to i. [3 pts: 1 per equation, 1 for the two meanings]

(b) *Invariance.* Every column of P sums to 1 ($0 + \frac{1}{2} + \frac{1}{2}$) and every column of R sums to 1 ($0 + \frac{1}{3} + \frac{2}{3}$), so $\sum_i \pi_i P_{ij} = \frac{1}{3} \cdot 1 = \pi_j$ for both: π is invariant for P and for R. *Detailed balance.* P: $\pi_i P_{ij} = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6} = \pi_j P_{ji}$ for all three pairs $(0,1), (1,2), (0,2)$: holds. R: $\pi_0 R_{01} = \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$ but $\pi_1 R_{10} = \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$: fails. [3 pts: 1 invariance, 1 each for P and R]

(c) Sum $\pi_i P_{ij} = \pi_j P_{ji}$ over i: $\sum_i \pi_i P_{ij} = \pi_j \sum_i P_{ji} = \pi_j \cdot 1 = \pi_j$, which is invariance. The converse fails for R: π is invariant but detailed balance does not hold, so invariance is the weaker statement. [2 pts]

(d) Mass from 0 to 1: $\pi_0 R_{01} = \frac{2}{9}$; from 1 to 0: $\pi_1 R_{10} = \frac{1}{9}$. The law stays π, but there is a net flow of $\frac{1}{9}$ around the cycle $0 \to 1 \to 2 \to 0$. Played backwards, the film shows a walker circulating the other way ($0 \to 2 \to 1 \to 0$ twice as often as the reverse), which the forward R-walker never does: the backwards film is distinguishable. For P each edge carries equal traffic in both directions ($\frac{1}{6}$ each), so the backwards film has the same statistics as the forward one; that is why detailed balance is also called reversibility. [2 pts: 1 for the two flows, 1 for the description]

Scoring: (a) 3, (b) 3, (c) 2, (d) 2. Partial credit for a correct method with an arithmetic slip.